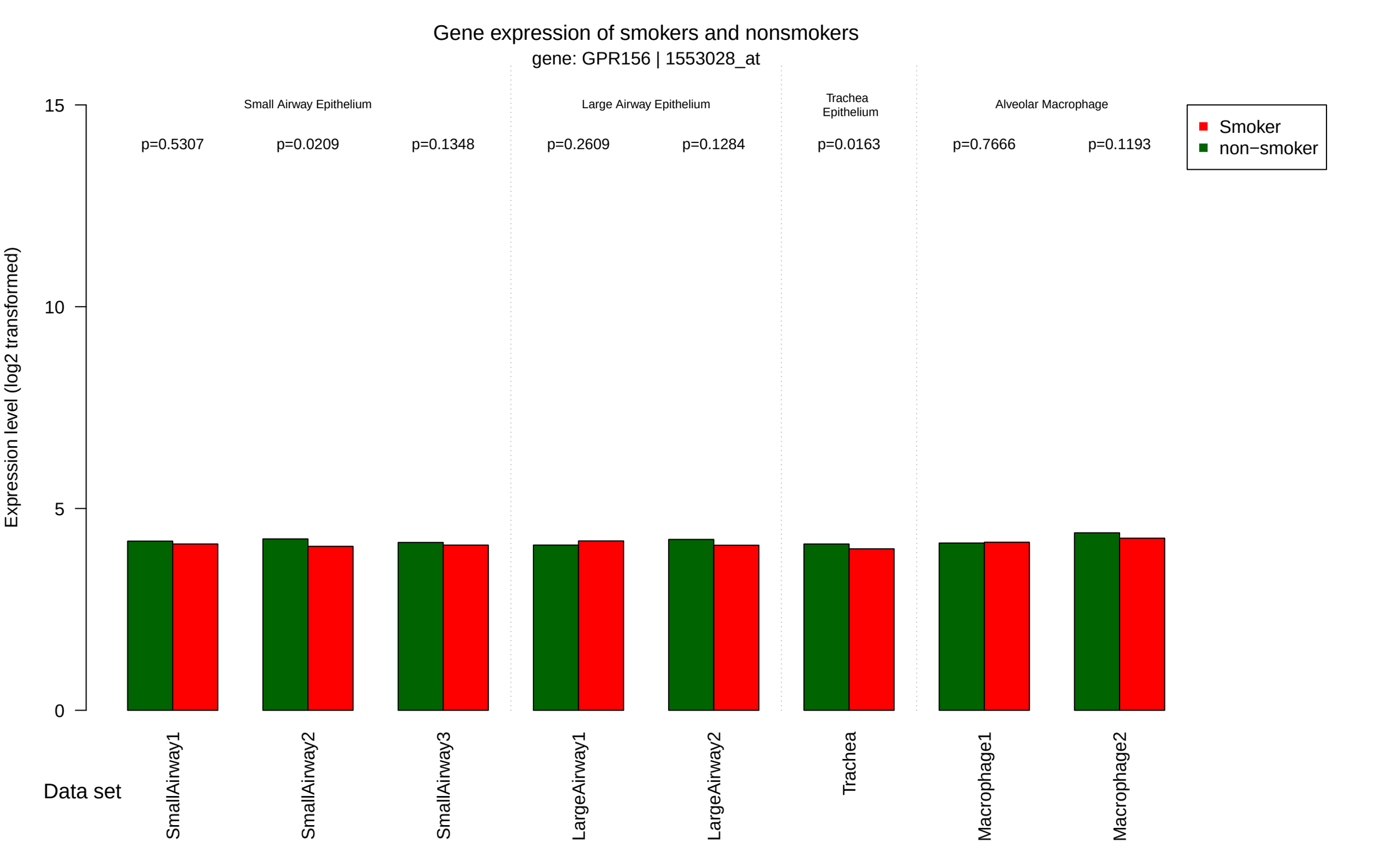
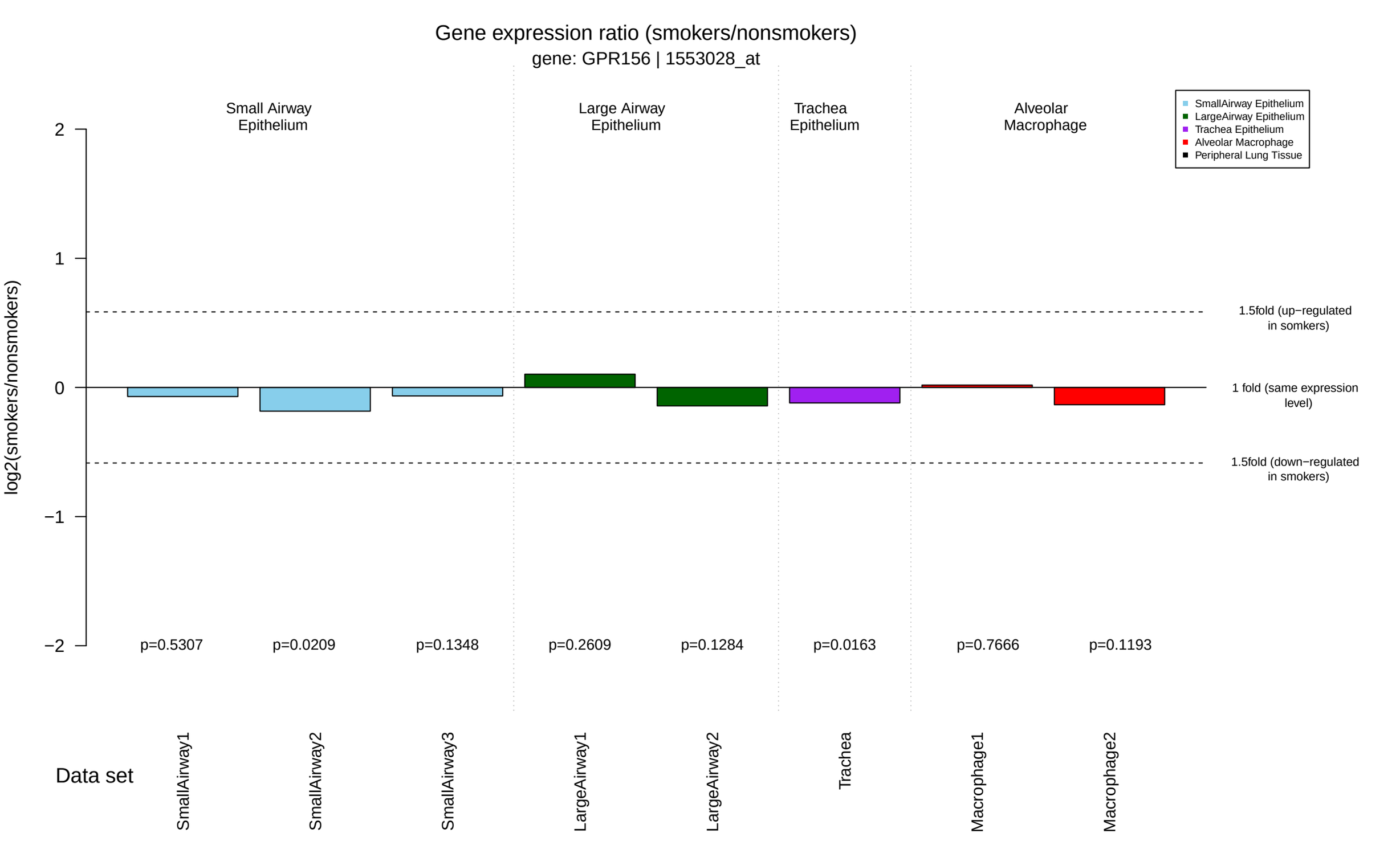
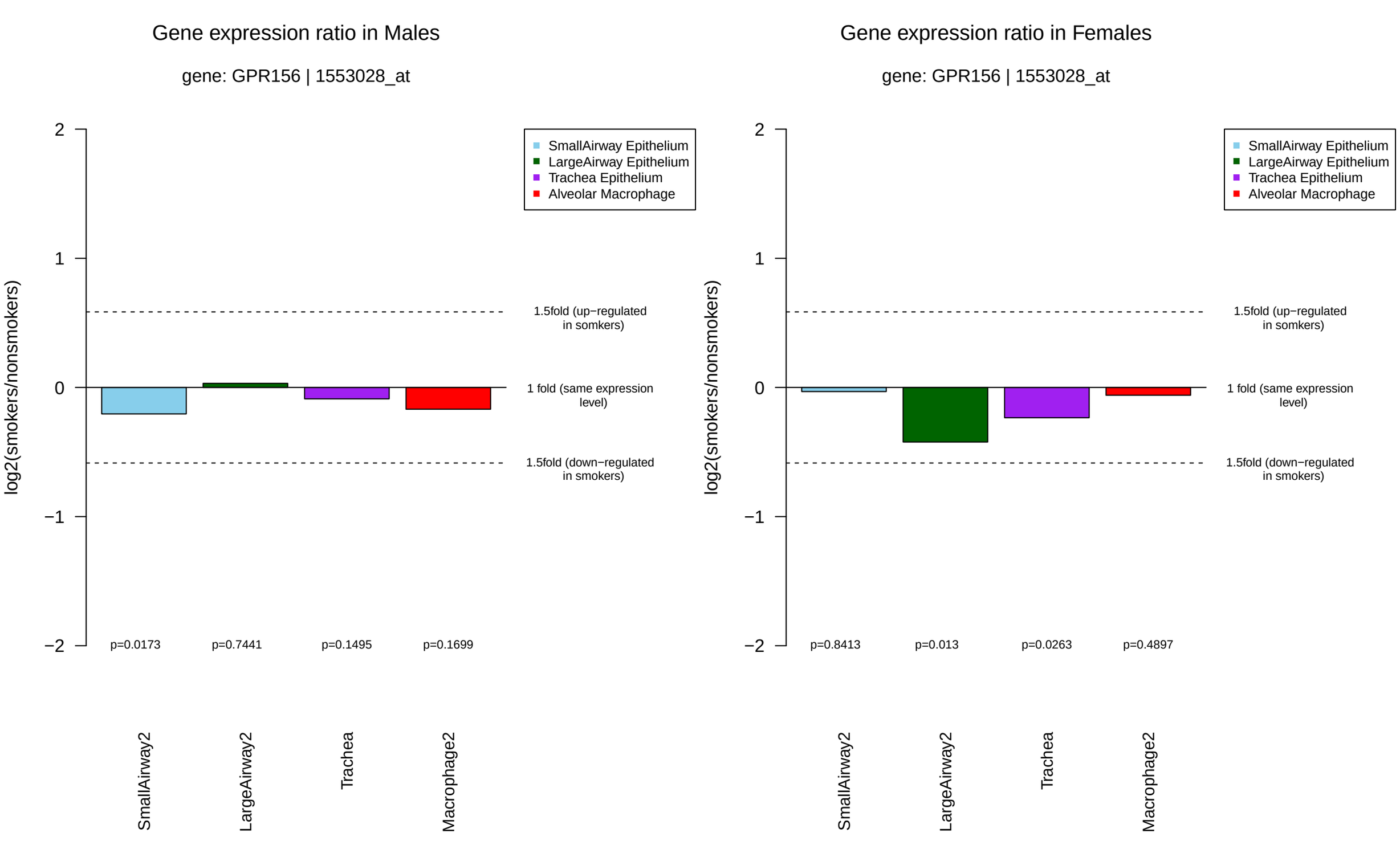
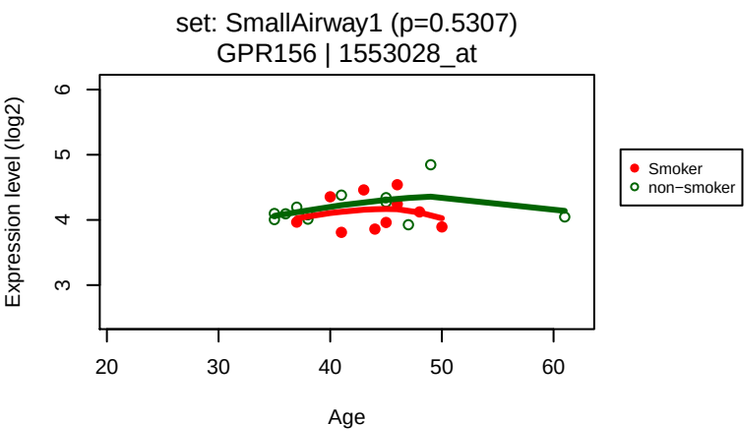
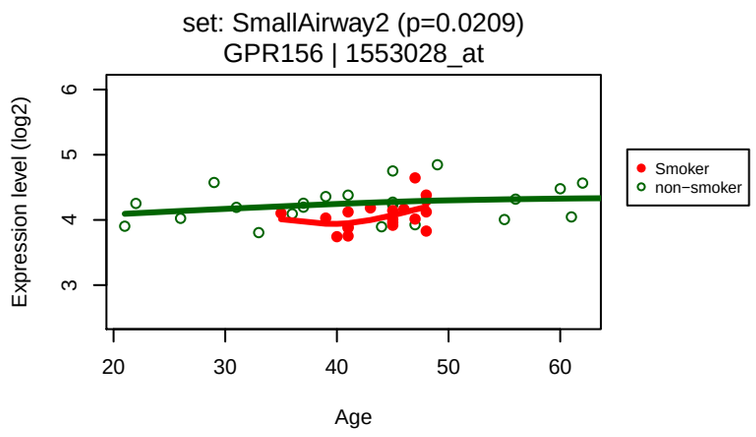
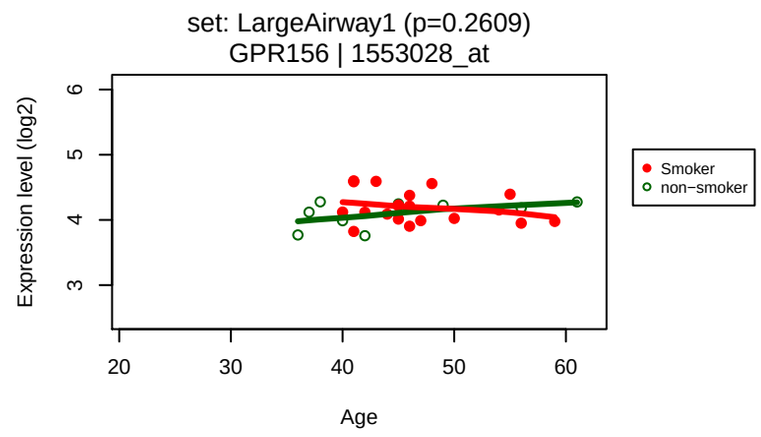

Gene expression of smokers and nonsmokers
gene: GPR156 | 1553028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5307
p=0.0209
p=0.1348
p=0.2609
p=0.1284
p=0.0163
p=0.7666
p=0.1193
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

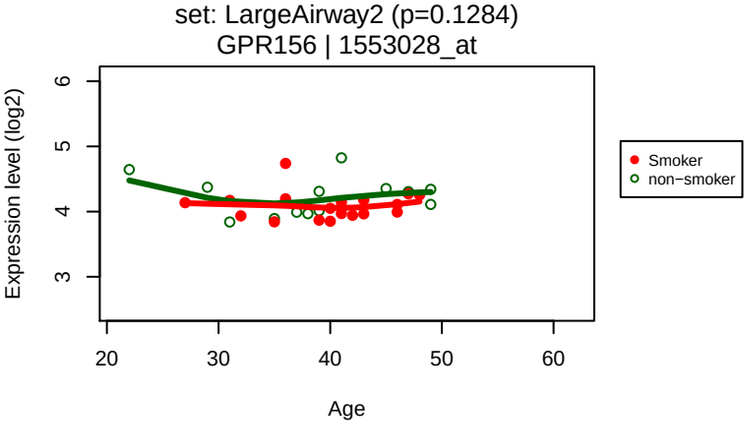

Gene expression ratio (smokers/nonsmokers)
gene: GPR156 | 1553028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5307
p=0.0209
p=0.1348
p=0.2609
p=0.1284
p=0.0163
p=0.7666
p=0.1193
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

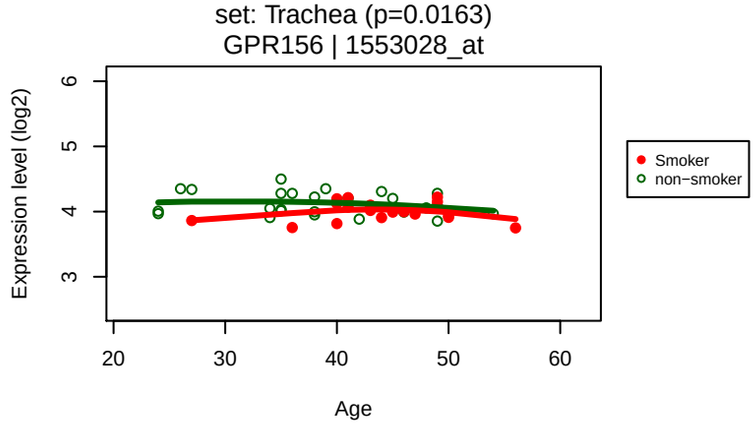

Gene expression ratio in Males
gene: GPR156 | 1553028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.7441
p=0.1495
p=0.1699
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR156 | 1553028_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8413
p=0.013
p=0.0263
p=0.4897
SmallAirway2
LargeAirway2
Trachea
Macrophage2

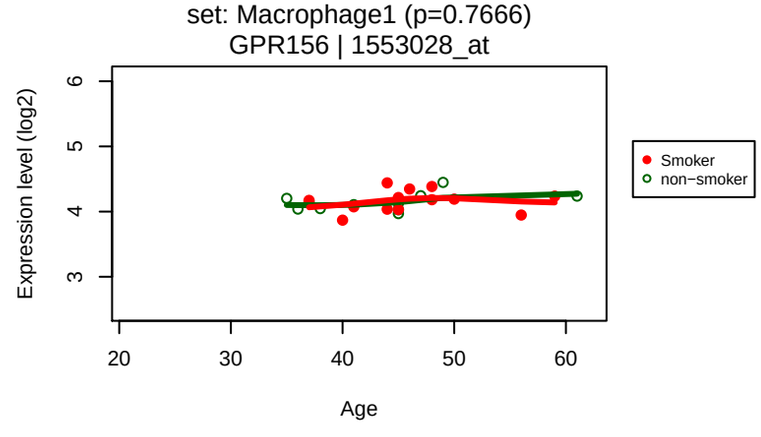

set: SmallAirway1 (p=0.5307)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker

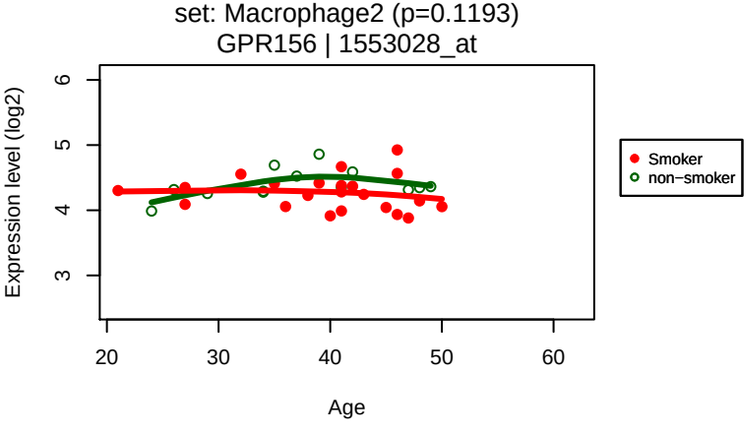

set: SmallAirway2 (p=0.0209)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker

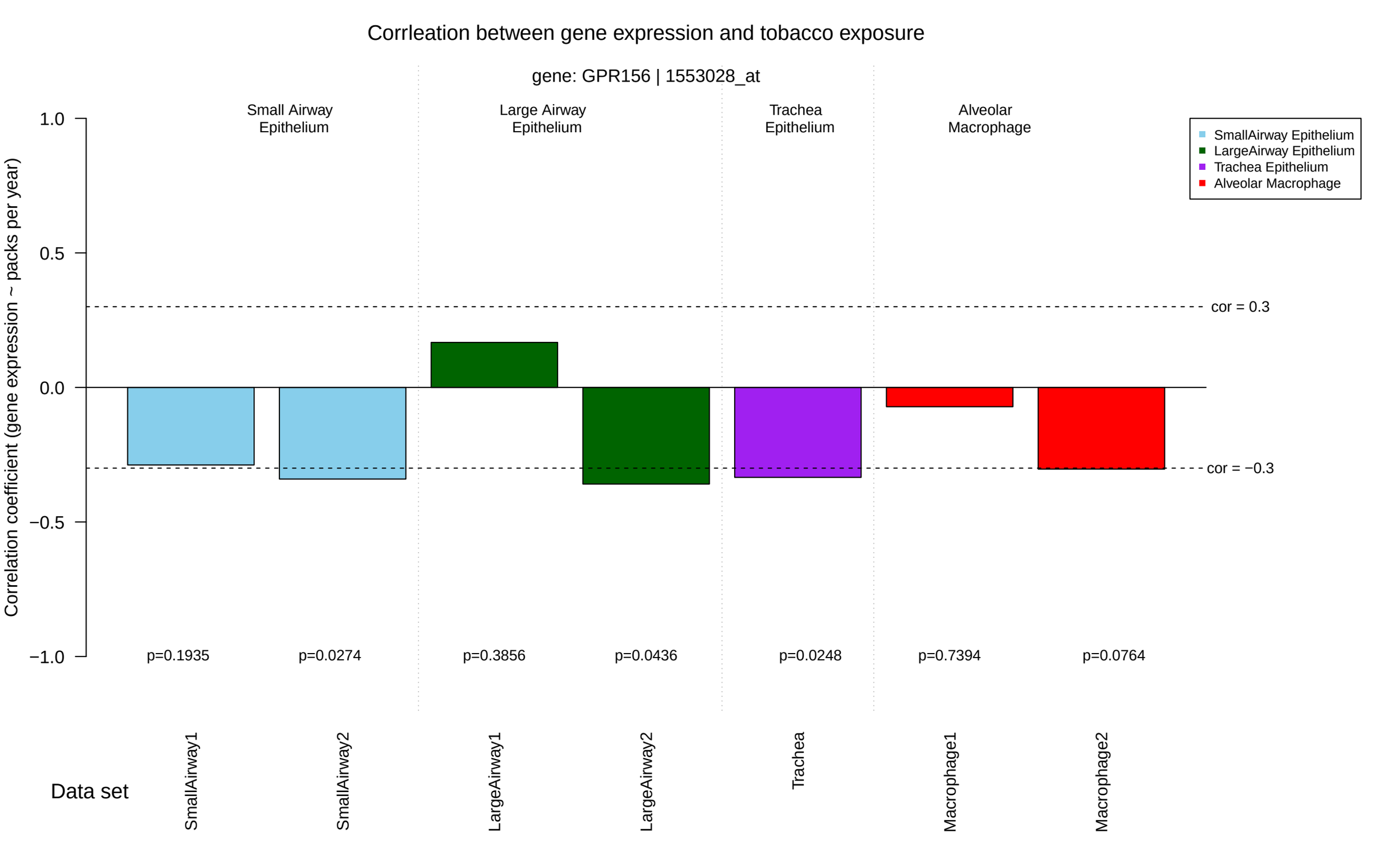

set: LargeAirway1 (p=0.2609)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1284)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0163)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7666)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1193)
GPR156 | 1553028_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR156 | 1553028_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1935
p=0.0274
p=0.3856
p=0.0436
p=0.0248
p=0.7394
p=0.0764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2